BCSE497J - PROJECT-I

A RELIABILITY-GUIDED VIT–MAMBA FRAMEWORK WITH FACIAL LANDMARK FUSION FOR STUDENT ENGAGEMENT RECOGNITION

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ABSTRACT

Recognising how engaged a student is from nothing more than a webcam feed is hard for two separate reasons. The classifier itself is weak, and the frames it is given are often not worth much. Someone leans out of shot, props their chin on a hand, or the camera drifts out of focus, and the model is still expected to read an emotional state off a face it can barely see. The usual fix is to throw those frames away. That cleans up the image quality, but it also wrecks the even spacing in time that a sequence model relies on.

In this project we take a different route on the DAiSEE dataset: no sampled frame is ever removed, each one is simply weighted. For every frame we compute a Motion Reliability Score (MRS) out of five signals that describe how good the observation is (blur, whether the face is visible, head pose, eye and iris visibility, and motion consistency). None of these touches the engagement label. A frozen Vision Transformer pretrained on ImageNet turns each aligned face crop into a 768-dimensional embedding. The reliability score scales that embedding and also weights the pooling step over time, so a poor frame carries less influence while staying exactly where it is in the sequence. A Mamba selective state-space model then reads the weighted sequence. Alongside this, a MediaPipe branch contributes 28 normalised geometric measurements and 52 ARKit blendshape values per frame, and the two branches are joined before a four-way classifier.

Instead of pinning the five MRS weights to some arbitrary equal value, we let the model adjust them as training proceeds, using what it sees in each epoch. They start at 0.20 apiece and move by gradient descent. The pipeline reaches 38.75% validation accuracy where chance would give 25% (binomial $p = 0.0046$), and SHAP attribution shows which groups of facial features the classifier is actually leaning on.

Keywords - Student Engagement Recognition, Vision Transformer, Mamba State-Space Model, Observation Reliability, MediaPipe Face Landmarks, DAiSEE, Explainable AI.

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1. INTRODUCTION

1.1 Background

Few things predict how much a student learns better than how engaged they are, and measuring that has become a pressing problem now that so much teaching happens online. A teacher standing in a room picks up on the mood constantly and adjusts on the fly. On a video call, with faces tiled in a grid, most of that signal is gone. Automatic engagement recognition from webcam footage is an attempt to bring it back.

For this task the usual public benchmark is DAiSEE. It holds 9,068 clips of ten seconds each, gathered from 112 people in everyday surroundings such as hostel rooms, labs and offices. Every clip carries labels for four affective states (boredom, confusion, engagement and frustration), each rated on four levels of intensity. We work with the engagement label alone.

Engagement recognition differs from ordinary facial expression recognition in one important way: engagement lives in time. You cannot see it in a single still. It shows up in how long attention holds, in where the gaze wanders, in how the body settles over a stretch of seconds. So the model has to read a sequence, not a frame. State-space models like Mamba have recently made this practical, since they handle sequences in linear rather than quadratic time.

1.2 Motivation

What pushed us towards this project is a weakness in existing pipelines that gets little attention: they assume every frame can be trusted equally. Real webcam recordings are nothing like that. The person leans out of view, a hand drifts over the mouth, the autofocus hunts, the lighting drops. A degraded frame yields degraded visual evidence, and handing that to the classifier at full weight simply pushes noise into the prediction.

The standard answer is gating: score the frame quality, then drop anything under a threshold. For a temporal model this is exactly the wrong move. Once frames are removed the sequence gets shorter, the gaps close up, and the even spacing that carried the temporal information in the first place is gone. A heavily occluded clip and a clean one now reach the sequence model with different lengths, and the model has no way to tell that one of them has lost time.

Our approach handles the same signal differently. Reliability should act as a soft weight applied in place rather than as a hard gate that deletes. Every sampled frame keeps its slot and its timing; the unreliable ones just count for less. The state-space model gets what it needs, and bad observations still cannot take over the prediction.

1.3 Scope of the Project

The project spans the whole recognition pipeline, starting from raw DAiSEE video and ending with a four-class engagement prediction plus an explanation of it. Inside the scope are: uniform

sampling over time; detecting, aligning and cropping the face; computing a five-part Motion Reliability Score; extracting features with a frozen Vision Transformer; reliability-weighted temporal modelling using a Mamba selective state-space model; a parallel MediaPipe branch for geometric and behavioural features; fusing the branches and classifying; and SHAP attribution over the fused representation.

Only the engagement label is predicted. Boredom, confusion and frustration exist in the dataset but are left alone. The Vision Transformer stays frozen the whole time; fine-tuning it end to end has been put off on purpose until the pipeline itself has been shown to work. Evaluation uses the official DAiSEE split, and we check that no participant appears in more than one split rather than taking it on trust.

We are not setting out to beat the best published DAiSEE accuracy, and the report should not be read that way. The question we care about is whether guiding the temporal model with reliability makes recognition more robust without losing temporal information. What we deliver is a pipeline that has been checked and can be rerun, with results reported as measured, negative ones included.

2. PROJECT DESCRIPTION AND GOALS

2.1 Literature Review

The relevant prior work splits into five threads: the dataset and benchmark, temporal architectures for video, state-space sequence models, facial feature extraction, and ways of judging observation quality.

Dataset and benchmark

DAiSEE was introduced by Gupta et al. [1] as the first sizeable dataset for spotting affective states in e-learning from unconstrained webcam video, and it set up the four-state, four-level labelling scheme we rely on. Their baselines, which used InceptionNet features per frame along with C3D and LRCN, made clear how hard the benchmark is, especially on the rarer intensity levels. Later studies keep finding the same thing: the engagement labels in DAiSEE are badly skewed towards the two middle levels, so reporting accuracy on its own says little [2], [3].

Temporal modelling for engagement

Since engagement plays out over time, most of the stronger methods model sequences instead of individual frames. People have tried LSTMs over per-frame CNN descriptors [4], 3D convolutional networks that learn spatial and temporal filters together [5], and more recently video transformers that run self-attention across frames [6]. Attention-based models have tended to beat purely recurrent ones here, but self-attention grows quadratically with sequence length, which caps how many frames can be handled.

State-space sequence models

Mamba, from Gu and Dao [7], is a selective state-space model whose parameters depend on the input, so information moves along the sequence in a content-aware way while staying linear in time and constant in inference memory. Zhu et al. [8] brought the idea to images as Vision Mamba (Vim), stacking bidirectional Mamba blocks over patch tokens to form a backbone. The two uses should not be confused: Vim is a spatial backbone over image tokens, while running Mamba over a series of per-frame embeddings is temporal modelling. This project does the second and describes it as such. Mamba variants have already been tried on DAiSEE; a recent multi-scale contrastive framework reports Vision Mamba and VMamba baselines there [9], so we make no claim to be the first to apply Mamba to engagement recognition.

Facial feature extraction

Lugaresi et al. [10] present MediaPipe, whose Face Mesh model predicts 468 three-dimensional facial landmarks in real time on ordinary hardware; iris refinement later took this to 478 points [11]. Kartynnik et al. [12] describe the mesh regression underneath. On top of raw coordinates, the MediaPipe Face Landmarker also outputs 52 ARKit-style blendshape activations, semantic descriptors like eye blink, brow lowering and jaw opening. These sit close in spirit to Facial Action Units [13] but come without a separate AU model. Raw landmark coordinates are swamped by

face scale and position, so normalised geometric quantities such as the eye aspect ratio [14] are the usual choice.

Observation quality and reliability

Estimating image sharpness is an old problem; the variance of the Laplacian [15] is still the go-to no-reference measure even though it is known to react to contrast, which has prompted perceptual alternatives [16]. For faces in particular, image quality assessment has been shown to predict recognition error [17], which means a quality estimate can reasonably be used to temper downstream confidence. Weighting inputs by uncertainty or confidence is well trodden ground in multimodal learning [18], and attention-based frame weighting has been used for video face recognition [19]. What is much less common is a pipeline that computes an explicit, interpretable, label-free reliability score per frame and uses it to modulate a temporal sequence model without deleting anything. That is the gap we go after.

Explainability

SHAP, introduced by Lundberg and Lee [20], is a unified additive attribution framework with some attractive theoretical guarantees, and it has become the default for post-hoc feature attribution in affective computing. We use it to work out which groups of facial features the classifier depends on, with the obvious caveat that attribution does not amount to causation.

2.2 Research Gap

Reading the literature above, three gaps stand out.

- **Reliability is used as a gate rather than a weight.** Quality-aware pipelines today usually put a threshold on frame quality and drop whatever falls under it. For a temporal model that defeats the purpose: taking frames out changes the sequence length and breaks the even spacing, so a degraded clip and a clean one stop being comparable inputs. As far as we can find, no widely reported DAiSEE pipeline keeps the full temporal sequence while turning down unreliable frames in place.
- **Reliability weights are asserted, not learned.** When a multi-factor quality score is used at all, the relative weight of the factors is nearly always set by hand, and usually to equal values. Whether blur, occlusion, head rotation and motion really deserve the same say for a particular corpus and camera setup is an empirical question. It is seldom asked, and it is never settled by optimisation.
- **Reliability and behaviour are conflated.** Head rotation and closed eyes are routinely read as signs of disengagement. That mixes up a statement about observation quality with the thing being predicted: a turned head only means the frontal appearance evidence is weaker, and a blink is ordinary behaviour, not a bad observation. Once a reliability score is contaminated by the label, it stops being a reliability score.

2.3 Objectives

The project sets out to achieve the following objectives, each of which can be checked.

- **O1.** Build a DAiSEE processing pipeline in which the official train/validation/test partition is used exactly as released and the overlap of participants between the three splits is measured and shown to be zero.
- **O2.** Define and implement a five-part Motion Reliability Score in [0,1] that draws only on observation-quality evidence, with every component checked to be finite and within bounds across the whole corpus.
- **O3.** Bring the reliability score into a ViT-plus-Mamba temporal architecture in a way that provably keeps the number and order of temporal positions intact, so that no frame is ever dropped.
- **O4.** Swap the arbitrary equal weighting of the reliability components for weights learned by gradient descent, starting from 0.20 each, and log how they move over training.
- **O5.** Fuse a MediaPipe geometric and behavioural branch with the deep temporal branch, fitting every dataset-dependent normalisation statistic on the training split only.
- **O6.** Report accuracy, weighted F1, macro F1, per-class precision/recall/F1 and a confusion matrix, set against explicit chance and majority-class baselines, with statistical significance stated.
- **O7.** Produce SHAP attribution over the fused representation, grouped into facial feature families that a person can interpret.

2.4 Problem Statement

Take a ten-second webcam clip of one learner and predict the labelled engagement intensity as one of four ordered levels (Very Low, Low, High, Very High), knowing that the quality of the frames inside the clip varies and that this has to be dealt with without throwing temporal information away.

Put more formally: a clip gives T uniformly sampled frames. For each frame t, a reliability score r_t in [0,1] is computed purely from observation-quality evidence, and a frozen Vision Transformer yields an embedding f_t. The system has to learn a mapping from the reliability-weighted sequence to a four-class prediction such that every clip has the same sequence length and the same ordering of temporal positions no matter how degraded its frames are, and such that the reliability score never depends on the engagement label.

Three things about the data make this harder. The classes in the full DAiSEE release are badly imbalanced. Neighbouring intensity levels like High and Very High look alike, even to the people who labelled them. And a single participant turns up in many clips, so a model can look good simply by learning who someone is rather than how engaged they are, unless the splits keep participants apart.

2.5 Project Plan

Project-I was broken into eleven tasks over roughly sixteen weeks, ordered so that each checkpoint comes before the work that depends on it. Auditing the dataset comes before any preprocessing; the reliability score is designed and calibrated before it goes into the model; and the temporal

encoder is checked against an independent reference implementation before any training starts. Experiments, the explainability analysis and the write-up take up the final stretch.

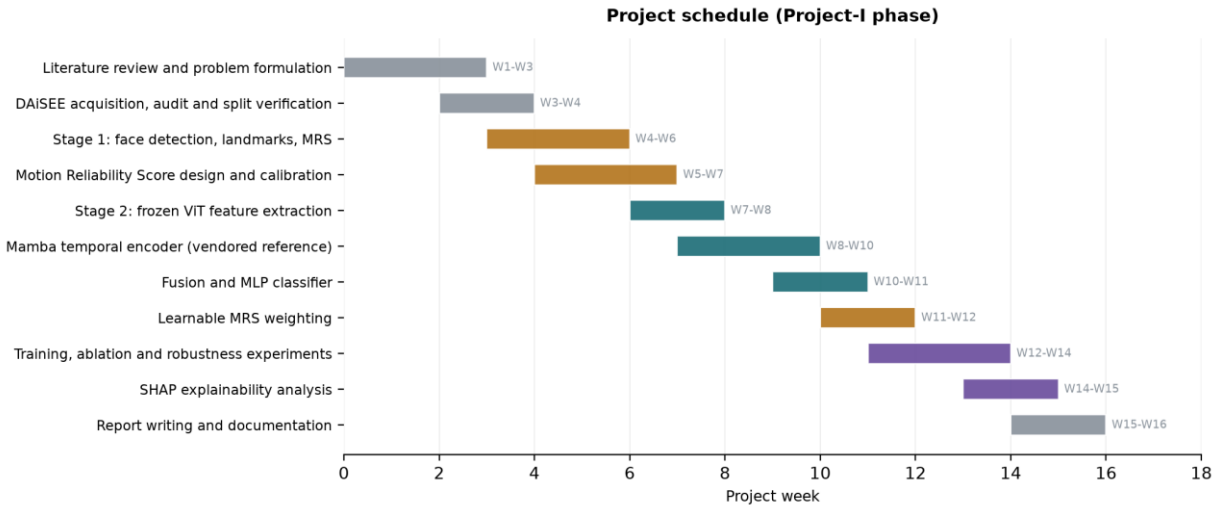


Fig. 2.1 Gantt chart of the Project-I schedule.

3. TECHNICAL SPECIFICATION

3.1 Requirements

3.1.1 Functional

The functional requirements are grouped by pipeline stage below, each phrased as something the system can be checked against.

- **Dataset ingestion and auditing (FR-1 to FR-5):** the system must find the DAiSEE directory tree, parse the official label CSV files, match every video file to its label row, and flag any label row with no file or any file with no label. It must decode-probe each video and record frame rate, frame count, resolution and duration as actually measured, not as assumed. It must work out the participant identifier for every clip and compute the pairwise subject overlap across train, validation and test, and refuse to go further if any overlap is non-zero.
- **Temporal sampling (FR-6 to FR-8):** the system must pick T frames per clip by uniform sampling, with T configurable over {8, 16, 32, 64}; sampling must be deterministic rather than random so that runs can be reproduced; and exactly T frames must come back for every clip, even clips shorter than T.
- **Face processing (FR-9 to FR-12):** the system must detect the participant's face in each sampled frame, roll-align the crop using the detected eye keypoints, pad and resize it to the backbone's native input size, and store the detector confidence. When no face is found it must fall back through an explicit, logged chain instead of dropping the frame, and must flag that frame as unreliable.
- **Reliability scoring (FR-13 to FR-17):** the system must compute five observation-quality components per frame (blur, face visibility, head pose, eye and iris visibility, and motion consistency), each within [0,1]; combine them into a single MRS value; fit any dataset-dependent calibration on the training split alone; and guarantee that the engagement label plays no part in the computation.
- **Feature extraction (FR-18 to FR-21):** the system must encode each aligned crop with a frozen pretrained Vision Transformer, reading the embedding dimension off the loaded model rather than hard-coding it; and must extract 478 facial landmarks, 52 blendshape activations and a head-pose matrix per frame, turning them into a fixed-width, scale-normalised feature vector.
- **Temporal modelling and classification (FR-22 to FR-26):** the system must apply the reliability score to the frame embeddings without changing the number or order of temporal positions; model the resulting sequence with a Mamba selective state-space encoder; pool the sequence into a clip embedding; fuse it with the landmark branch after separate normalisation and projection; and emit four raw class logits.
- **Learning and evaluation (FR-27 to FR-32):** the system must start the five reliability weights at 0.20 each and update them by gradient descent during training, saving the learned values in the checkpoint so they are reused at validation and test time; must choose the

reported checkpoint on validation macro F1 and never on test performance; and must report accuracy, weighted F1, macro F1, per-class metrics and a confusion matrix, along with SHAP attribution over the fused representation.

- **Caching and reproducibility (FR-33 to FR-36):** the system must cache expensive preprocessing under a key built from a digest of every setting that affects the cached content, so that a stale cache can never be picked up silently; and must record the exact package versions, device and configuration next to every result artifact.

3.1.2 Non-Functional

- **Correctness and verifiability:** every stage of the pipeline must have an automated check that fails loudly instead of quietly degrading. The temporal encoder in particular must be checked for causality and for numerical agreement with a separately written reference implementation of the underlying recurrence.
- **Data integrity and leakage prevention:** no statistic used at inference time, whether reliability calibration, feature scaler or class weights, may be fitted on validation or test data, and this must be enforced by the structure of the code rather than by convention.
- **Performance:** preprocessing the 216-clip development corpus must finish in under five minutes on a single consumer GPU, and a training epoch on cached features must take under two seconds, so that experiments can be turned around quickly.
- **Scalability:** the architecture must scale to the full 9,068-clip DAiSEE release through configuration changes alone, with preprocessing split so that changing the visual backbone does not mean redoing the CPU-bound landmark extraction.
- **Reproducibility:** all randomness must be seeded, and every reported figure must trace back to a committed artifact file, not to a console transcript.
- **Privacy and licensing:** the DAiSEE video, the face crops derived from it and the extracted biometric landmark data must never go into version control, and per-clip ground-truth labels must not be redistributed.
- **Maintainability:** sampling, face processing, reliability scoring, encoding, temporal modelling, fusion and evaluation must sit in separate modules, with no hard-coded dataset paths, tensor dimensions or model constants.

3.2 Feasibility Study

3.2.1 Technical Feasibility

- Every pretrained model we need is public and free: the ImageNet-pretrained ViT-B/16 backbone comes through the timm library, and the MediaPipe BlazeFace detector (230 KB) and Face Landmarker (3.76 MB) bundles are downloaded from Google's published model URLs. Nothing has to be pretrained from scratch.

- Keeping the Vision Transformer frozen brings the trainable parameters down from about 87 million to 1.37 million, which fits comfortably in the 6 GB available on the development GPU.
- One real technical risk came up during implementation and was dealt with. The official mamba-ssm package ships as source only and needs the CUDA Toolkit compiler to build, which the development machine did not have, so the install failed. Rather than swap in a different architecture, we vendored the official reference (non-fused) implementation from the Mamba repository, which computes the same function without any compilation. What we give up is speed, not correctness, and the backend in use is written into every result artifact.
- Splitting preprocessing into a CPU stage that does not depend on the backbone and a GPU stage for feature extraction keeps things cheap: swapping the visual backbone only costs the second stage, which runs in about 27 seconds on the development corpus.

3.2.2 Economic Feasibility

- There is no software cost. PyTorch, timm, MediaPipe, OpenCV, scikit-learn and SHAP are all open source, and the pretrained weights are released for research use.
- On the hardware side, everything runs on a single consumer laptop GPU the team already had. No cloud compute, paid APIs or annotation budget is involved.
- DAiSEE is provided free for research once a registration agreement is signed, so the data costs nothing to obtain.
- The main cost is our own time. Because features are cached, a complete training run finishes in under a minute, so the cost of running experiments is small next to the effort of building the pipeline.

3.2.3 Social Feasibility

- Pointing engagement recognition at students raises real ethical questions. We position the system as a research instrument that works on aggregates, not as a surveillance or grading tool, and we do not propose any decision about an individual student.
- The DAiSEE labels are outside observers' judgements of visible affect. They should not be mistaken for objective readings of what a learner is actually thinking or feeling, and the report says so plainly.
- The dataset is licence-restricted footage of identifiable people. For that reason the video, the face crops derived from it and the extracted landmark data are kept out of version control altogether, and per-clip label values are treated as something we cannot redistribute.
- Any use beyond research would need informed consent, ethics review by an institution, and evidence that performance does not differ across demographic groups. None of that is claimed here.

3.3 System Specification

3.3.1 Hardware Specification

Table 3.1 Hardware specification.

Component	**Development configuration (used)**	**Recommended for full DAiSEE**
Processor	x86-64, 8 logical cores	12+ cores (MediaPipe is CPU-bound)
Memory (RAM)	16 GB	32 GB
Storage	60 GB free (SSD)	250 GB SSD (15 GB dataset + caches)
Graphics Processing Unit	NVIDIA GeForce RTX 3060 Laptop, 6 GB, compute 8.6	NVIDIA RTX 3090 / A4000, 12-16 GB
Display	1920 x 1080	1920 x 1080 or higher

3.3.2 Software Specification

Table 3.2 Software specification (versions as verified on the development machine).

Layer	**Software**	**Version**
Operating system	Windows 11 Home (64-bit)	10.0.26200
Language	Python	3.13.3
Deep learning	PyTorch / torchvision	2.11.0+cu128 / 0.26.0+cu128
GPU runtime	CUDA (bundled with PyTorch wheel)	12.8
Vision backbones	timm (PyTorch Image Models)	1.0.29
Transformer hub	transformers	5.16.1
Facial analysis	MediaPipe Tasks (Face Detector, Face Landmarker)	1.0.1
Image processing	OpenCV	5.0.0
Sequence model	Vendored Mamba reference (state-spaces/mamba)	pytorch_reference
Numerics	NumPy	2.5.2
Metrics	scikit-learn	1.9.0
Explainability	SHAP	0.52.0
Plotting	Matplotlib	3.11.1
Testing	pytest	9.1.1

4. DESIGN APPROACH AND DETAILS

4.1 System Architecture

Fig. 4.1 System architecture of the MRG-ViT-Mamba engagement recognition pipeline

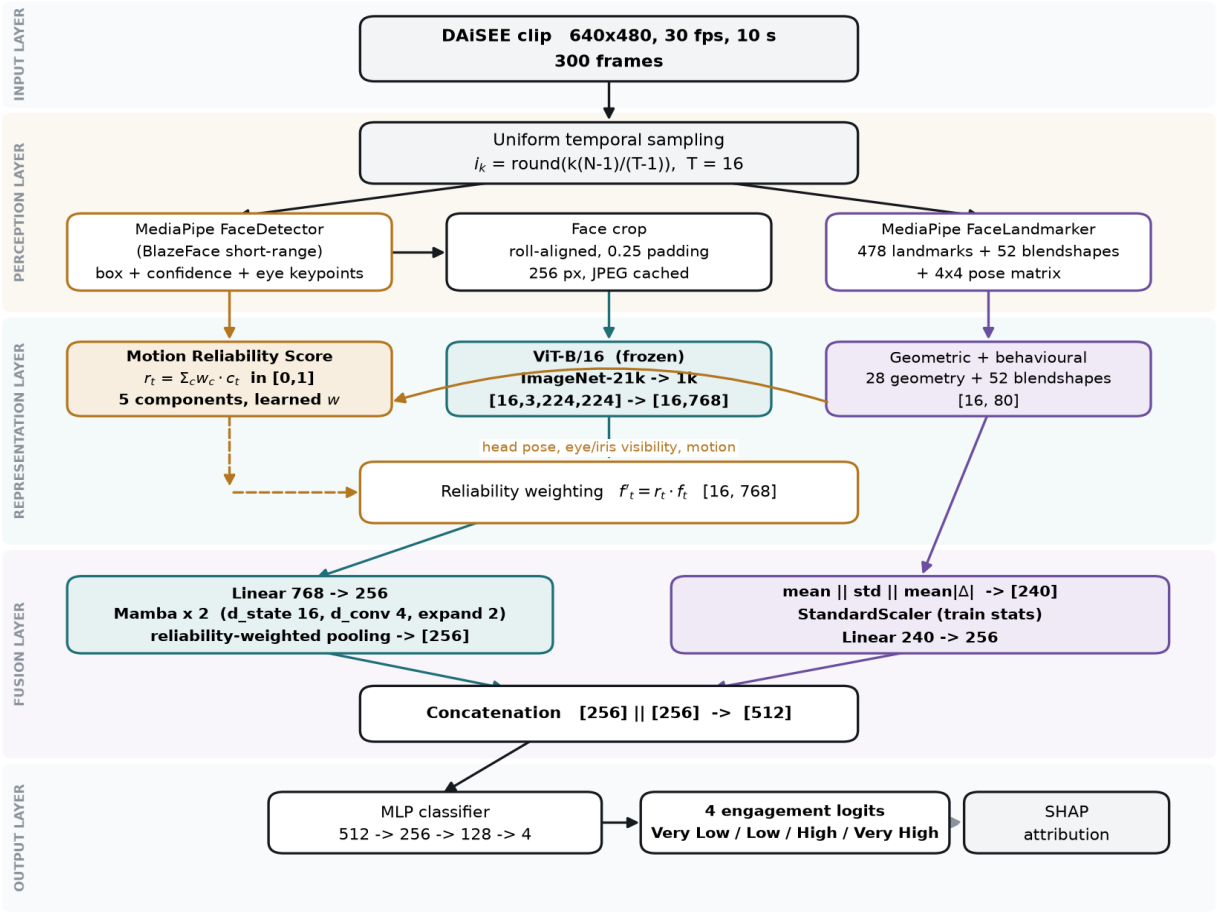


Fig. 4.1 System architecture of the MRG-ViT-Mamba engagement recognition pipeline.

The architecture has five layers. The input layer takes a ten-second DAiSEE clip at 640 x 480 and 30 frames per second, and the perception layer pulls T = 16 evenly spaced frames from it using the index rule i_k = round(k(N-1)/(T-1)). Because the sampling is deterministic, rerunning the pipeline gives back exactly the same frames.

In the perception layer, two pretrained MediaPipe models run on every sampled frame. The BlazeFace short-range detector gives a bounding box, a confidence score and six facial keypoints; the two eye keypoints roll-align the crop, which is then padded by 25 percent and resized. Separately, the Face Landmarker returns 478 three-dimensional landmarks (ten of them on the irises), 52 ARKit blendshape activations, and a 4 x 4 facial transformation matrix from which yaw, pitch and roll are recovered.

The representation layer is where the three parallel computations that define this design happen. The Motion Reliability Score folds five observation-quality signals into one value per frame. The

frozen ViT-B/16 maps each aligned crop to a 768-dimensional embedding; the batch is held as [B,T,C,H,W], reshaped to [B*T,C,H,W], and brought back to [B,T,768] in one call instead of a loop over frames. The landmark branch turns the raw mesh into 28 scale-normalised geometric measurements which, together with the 52 blendshapes, form an 80-dimensional descriptor per frame. Reliability weighting is then applied as f'_t = r_t * f_t, which by construction leaves the tensor shape and every temporal position untouched.

The fusion layer projects the reliability-weighted sequence from 768 down to 256 dimensions and runs it through two Mamba blocks. Each block consists of RMSNorm, an input projection, a causal depthwise convolution, a SiLU gate and the selective scan, wrapped in a residual connection. The sequence is then collapsed by reliability-weighted pooling, D = sum_t r_t h_t / sum_t r_t, and this is the point where reliability has the most say over the clip representation. Meanwhile the per-frame landmark descriptors are summarised over time by concatenating their mean, standard deviation and mean absolute temporal difference, producing a 240-dimensional clip vector that is standardised with training-split statistics and projected to 256 dimensions. The two 256-dimensional outputs are concatenated into a 512-dimensional vector.

The output layer is a multilayer perceptron going 512 to 256 to 128 to 4, with GELU activations and dropout, that produces four raw logits trained with class-weighted cross-entropy. SHAP attribution is computed on the representation just before fusion, where the landmark half still has named dimensions and can be grouped into recognisable facial families.

Table 4.1 Parameter budget of the assembled model.

Module	Detail	Parameters	Trained
ViT-B/16 backbone	frozen, preprocessing stage	85,798,656	No
Temporal input projection	Linear 768 -> 256	196,864	Yes
Mamba blocks	2 blocks, d_inner 512, d_state 16	876,032	Yes
Final RMSNorm	256	256	Yes
Deep branch projection	LayerNorm + Linear 256 -> 256	66,304	Yes
Landmark branch projection	Scaler + Linear 240 -> 256	61,696	Yes
Learnable MRS weights	softmax over 5 components	5	Yes
MLP classifier	512 -> 256 -> 128 -> 4	164,740	Yes
Trainable total		1,365,897	
Whole pipeline		87,164,553	

4.2 Design

4.2.1 Data Flow Diagram

Fig. 4.2 Level-0 data flow diagram — engagement inference on one clip

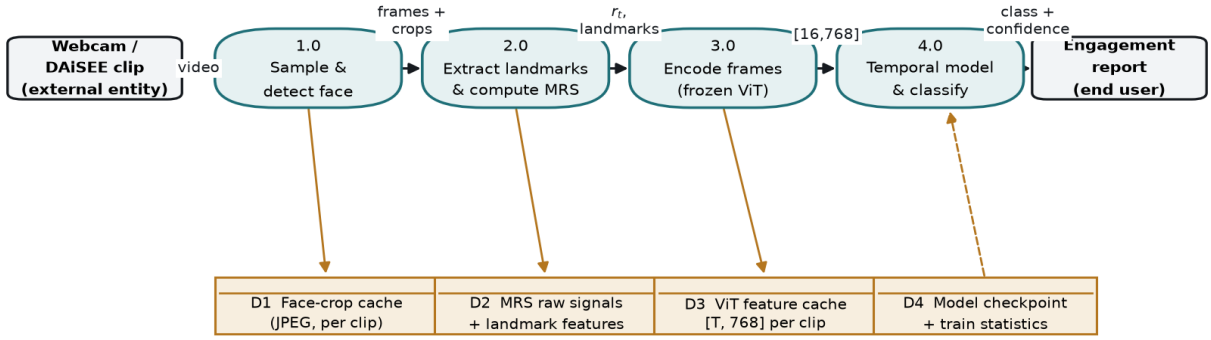


Fig. 4.2 Level-0 data flow diagram for engagement inference on a single clip.

The diagram follows a single clip through the system. It enters process 1.0, which samples sixteen frames, then detects, aligns and crops the face and writes the crops to data store D1. Process 2.0 extracts landmarks and blendshapes and computes the five raw reliability signals, storing them in D2 before any calibration. Keeping the raw signals rather than a finished score was a conscious choice: the reliability calibration can be refitted, or the component weights changed, without decoding a single video again. Process 3.0 encodes the cached crops with the frozen Vision Transformer and writes the per-frame embeddings to D3. Process 4.0 reads the cached embeddings, the reliability signals and the landmark descriptors, applies the reliability weighting, runs the Mamba temporal encoder and the fusion classifier, and outputs the predicted engagement level with a confidence. The dashed edge shows that process 4.0 also pulls the trained checkpoint and the training-split normalisation statistics from D4.

4.2.2 Use Case Diagram

Fig. 4.3 Use case diagram

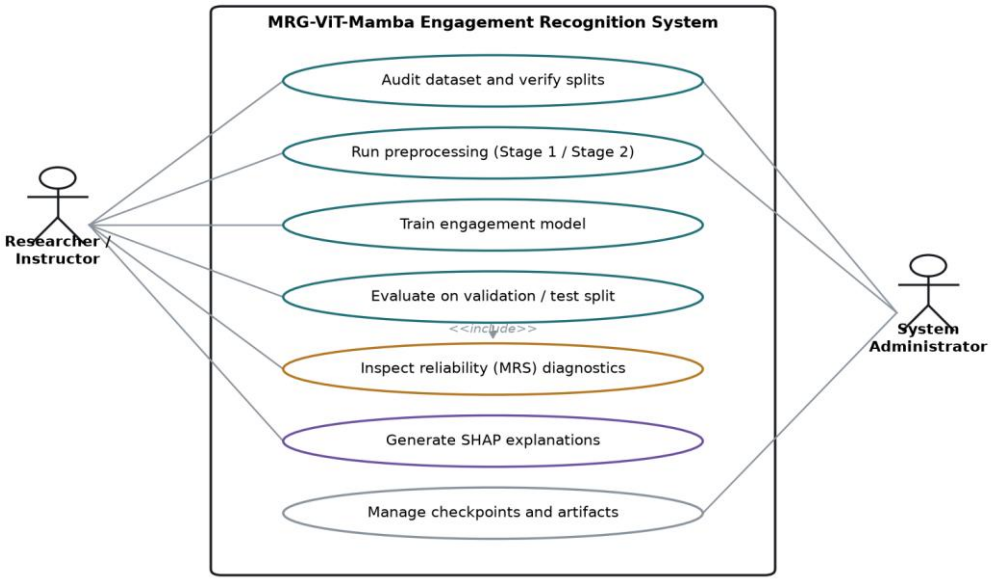


Fig. 4.3 Use case diagram.

There are two actors. The researcher or instructor audits the dataset and checks the splits, runs both preprocessing stages, trains the model, evaluates it on the validation and test partitions, looks over the reliability diagnostics, and generates SHAP explanations. Looking at the reliability diagnostics is part of evaluation because a per-class score only makes sense once you know how observable the underlying clips were. The system administrator handles checkpoints and artifacts, plus the preprocessing and audit operations, which are infrastructure rather than experiment.

4.2.3 Sequence Diagram

Fig. 4.4 Sequence diagram — engagement inference on a single clip

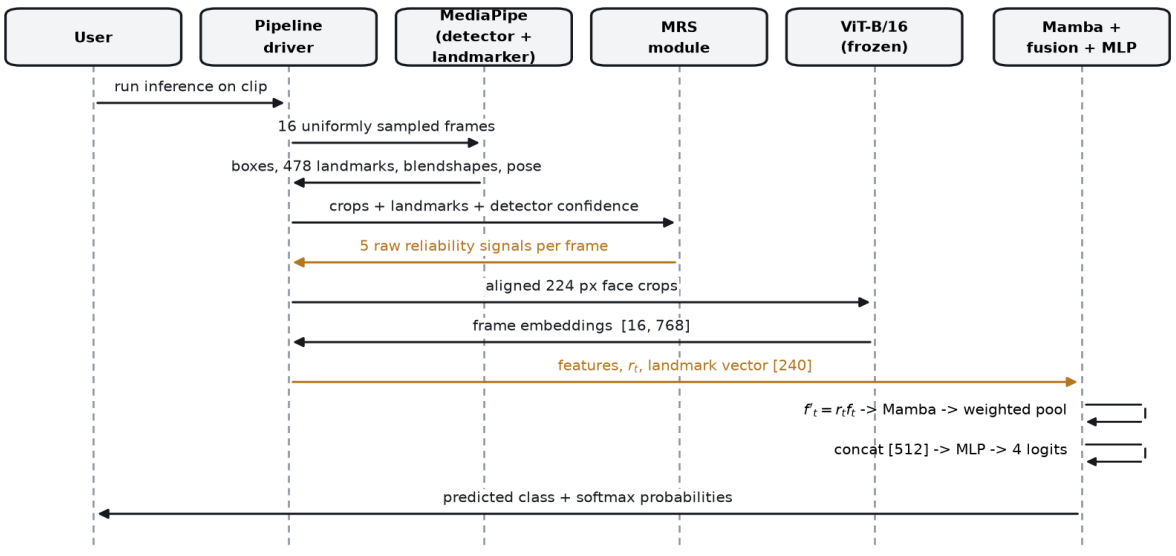


Fig. 4.4 Sequence diagram for engagement inference on a single clip.

The diagram lays out the order of calls during inference. The pipeline driver samples sixteen frames and hands them to MediaPipe, which sends back detector boxes, landmarks, blendshapes and pose matrices. These go to the reliability module along with the aligned crops and the detector confidence, and it returns the five raw signals for each frame. The crops are then encoded by the frozen Vision Transformer into a [16, 768] tensor. The temporal and fusion stage takes the embeddings, the reliability vector and the landmark descriptor, applies the weighting, runs the Mamba encoder and reliability-weighted pooling, concatenates the two branches and produces four logits, which come back as a predicted class with softmax probabilities.

5. REFERENCES

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Web resources:

DAiSEE dataset - https://people.iith.ac.in/vineethnb/resources/daisee/

Mamba reference implementation - https://github.com/state-spaces/mamba

Vision Mamba (Vim) - https://github.com/hustvl/Vim

MediaPipe Solutions - https://ai.google.dev/edge/mediapipe/solutions/guide

PyTorch Image Models (timm) - https://github.com/huggingface/pytorch-image-models